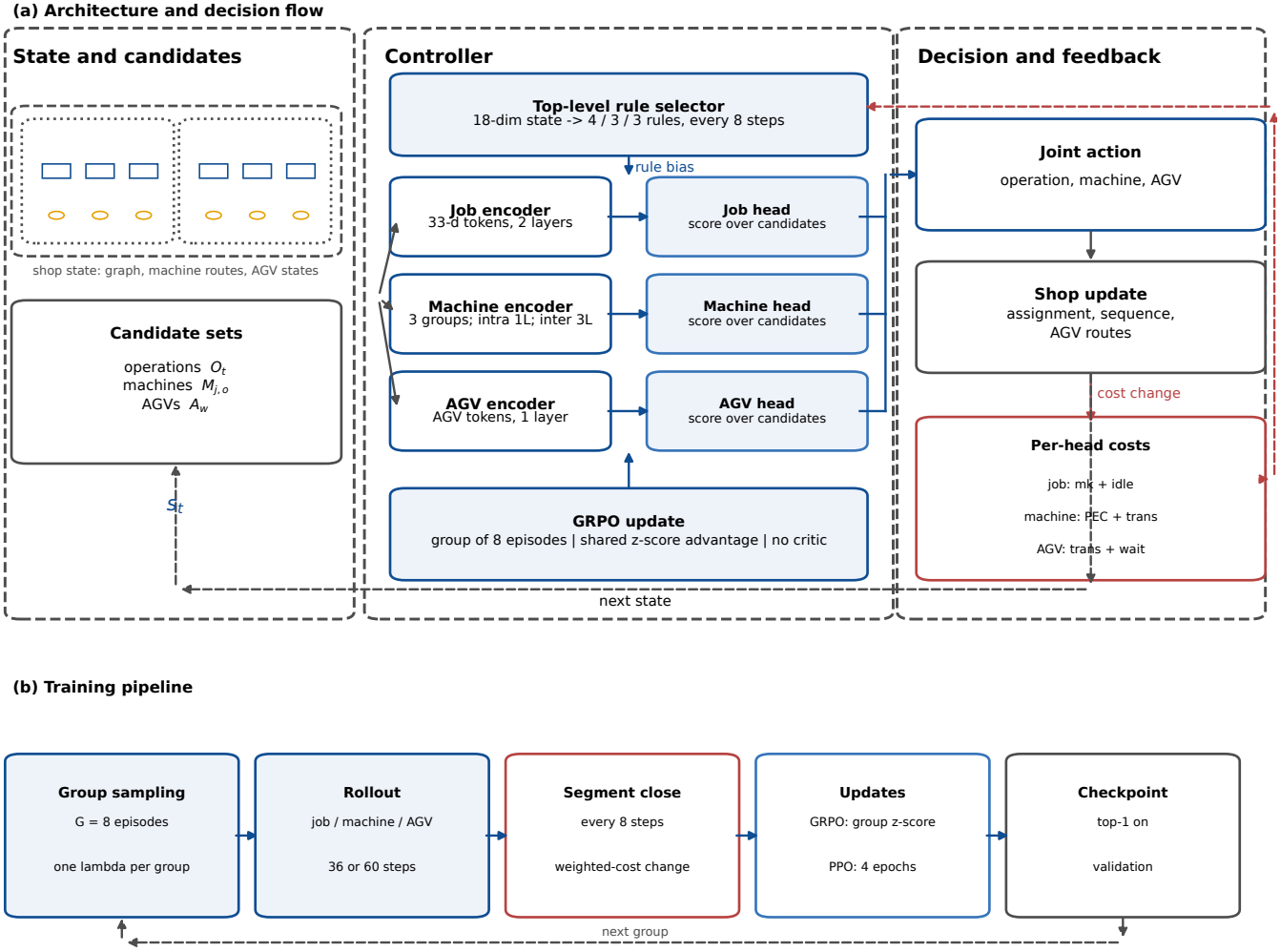

(a) Architecture and decision flow
State and candidates
shop state: graph, machine routes, AGV states
Candidate sets
operations O_t
machines $M_{j,o}$
AGVs A_w
S_t
Controller
Top-level rule selector
18-dim state -> 4 / 3 / 3 rules, every 8 steps
rule bias
Job encoder
33-d tokens, 2 layers
Job head
score over candidates
Machine encoder
3 groups; intra 1L; inter 3L
Machine head
score over candidates
AGV encoder
AGV tokens, 1 layer
AGV head
score over candidates
GRPO update
group of 8 episodes | shared z-score advantage | no critic
Decision and feedback
Joint action
operation, machine, AGV
Shop update
assignment, sequence,
AGV routes
cost change
Per-head costs
job: mk + idle
machine: PEC + trans
AGV: trans + wait
next state
(b) Training pipeline
Group sampling
G = 8 episodes
one lambda per group
Rollout
job / machine / AGV
36 or 60 steps
Segment close
every 8 steps
weighted-cost change
Updates
GRPO: group z-score
PPO: 4 epochs
Checkpoint
top-1 on
validation
next group